Optimality Conditions for Constrained Problems

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First-Order Optimality Condition

Theorem 1 (First-order necessary optimality condition)

Let $\Omega \subset \mathbb{R}^n$ be convex and $f : \mathbb{R}^n \to \mathbb{R}$ be differentiable. If $x \in \Omega$ is a minimizer of f over Ω, then

$$\langle \nabla f(x),\, y - x \rangle \geq 0, \qquad \forall y \in \Omega.$$

Optimality Condition for Convex Problems

Theorem 2

Let $\Omega \subset \mathbb{R}^n$ *be convex and* $f : \Omega \to \mathbb{R}$ *be convex and differentiable. Then*

$$-\nabla f(x) \in \mathcal{N}_\Omega(x) \quad \Longleftrightarrow \quad x \text{ minimizes } f \text{ over } \Omega.$$

Proof Idea

We already know

$$-\nabla f(x) \in \mathcal{N}_\Omega(x)$$

is necessary for optimality.

To prove sufficiency, note that

$$\langle \nabla f(x), y - x \rangle \geq 0 \quad \forall y \in \Omega.$$

By convexity of f,

$$f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle \geq f(x).$$

Therefore, x is a minimizer.

Normal Cone

Definition 3 (Normal cone)

Let $\Omega \subset \mathbb{R}^n$ be convex and let $x \in \Omega$. The normal cone to Ω at x, denoted $\mathcal{N}_\Omega(x)$, is defined as

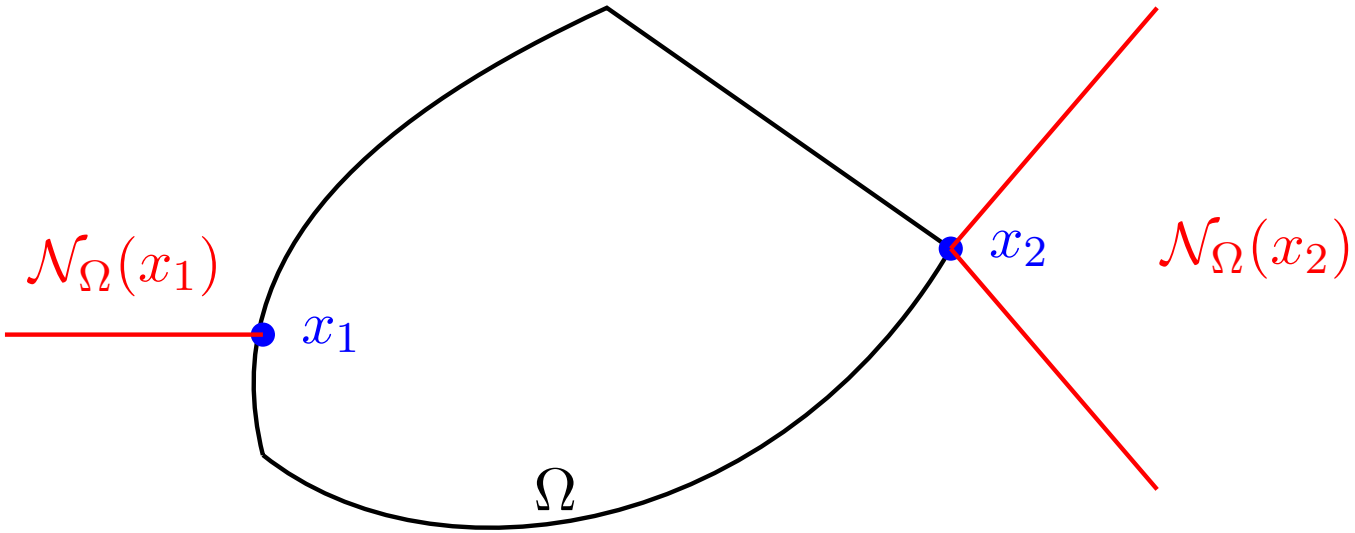
$$\mathcal{N}_\Omega(x) := \{d \in \mathbb{R}^n : \langle d, y - x \rangle \leq 0 \ \ \forall y \in \Omega\}.$$

With this definition, the first-order necessary optimality condition can be written as

$$-\nabla f(x) \in \mathcal{N}_\Omega(x).$$

Example: Normal Cones

As an example, here are normal cones computed for a convex set Ω.



Example: Normal Cone at an Interior Point

Example 4

Let Ω be convex and let x lie in the interior of Ω. Then the normal cone satisfies

$$\mathcal{N}_\Omega(x) = \{0\}.$$

Proof idea.

If some $d \neq 0$ belonged to $\mathcal{N}_\Omega(x)$, then for sufficiently small $\delta > 0$, the point $x + \delta d \in \Omega$ and

$$\langle d, x + \delta d - x \rangle = \delta \|d\|^2 > 0,$$

which contradicts the definition of the normal cone.

Therefore,

$$\mathcal{N}_\Omega(x) = \{0\}.$$

If x is an interior minimizer, then

$$\nabla f(x) = 0.$$

Normal Cone to a Hyperplane

Theorem 5

Let $\Omega := \{y \in \mathbb{R}^n : \langle a, y\rangle = 0\}$ *with* $a \neq 0$. *For any* $x \in \Omega$,

$$\mathcal{N}_\Omega(x) = \operatorname{span}\{a\} = \{\lambda a : \lambda \in \mathbb{R}\}.$$

Affine Plane Case

For an affine plane

$$\Omega := \{y \in \mathbb{R}^n : \langle a, y \rangle = b\},$$

the normal cone is unchanged:

$$\mathcal{N}_\Omega(x) = \operatorname{span}\{a\}, \qquad \forall x \in \Omega.$$

Affine Subspace Case

For the affine subspace

$$\Omega := \{y \in \mathbb{R}^n : Ay = b\},$$

the normal cone satisfies

$$\mathcal{N}_\Omega(x) = \operatorname{colspan}(A^\top), \qquad \forall x \in \Omega.$$

In particular, if $A = a^\top$, this reduces to

$$\mathcal{N}_\Omega(x) = \operatorname{span}\{a\}.$$

Lagrange Multipliers

Consider the constrained optimization problem

$$\min_x f(x) \quad \text{s.t. } Ax = b.$$

At optimality, the first-order condition implies

$$-\nabla f(x) = A^\top \lambda \quad \text{for some } \lambda \in \mathbb{R}^d.$$

The vector λ is called a Lagrange multiplier.

Projection onto an Affine Subspace

Let

$$\Omega = \{x \in \mathbb{R}^n : Ax = b\},$$

where $A \in \mathbb{R}^{d \times n}$ and $AA^\top$ is invertible.

We want the Euclidean projection of z onto Ω:

$$\min_{\boldsymbol{x}} \ \frac{1}{2}\|\boldsymbol{x} - \boldsymbol{z}\|_2^2 \quad \text{s.t. } \boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}.$$

Lemma 6

$$\boldsymbol{x} = \boldsymbol{z} - \boldsymbol{A}^\top(\boldsymbol{A}\boldsymbol{A}^\top)^{-1}(\boldsymbol{A}\boldsymbol{z} - \boldsymbol{b}).$$

Projection onto an Affine Subspace: Derivation

Gradient of the objective:

$$\nabla \tfrac{1}{2}\|\boldsymbol{x}-\boldsymbol{z}\|^2 = \boldsymbol{x}-\boldsymbol{z}.$$

First-order condition:

$$-(\boldsymbol{x}-\boldsymbol{z}) \in \mathcal{N}_\Omega(\boldsymbol{x}).$$

For $\Omega = \{\boldsymbol{x} : \boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}\}$,

$$\mathcal{N}_\Omega(\boldsymbol{x}) = \operatorname{colspan}(\boldsymbol{A}^\top).$$

Thus there exists $\boldsymbol{\lambda}$ such that

$$-(\boldsymbol{x}-\boldsymbol{z}) = \boldsymbol{A}^\top\boldsymbol{\lambda} \quad \Rightarrow \quad \boldsymbol{x} = \boldsymbol{z} - \boldsymbol{A}^\top\boldsymbol{\lambda}.$$

Using $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}$:

$$\boldsymbol{A}(\boldsymbol{z} - \boldsymbol{A}^\top\boldsymbol{\lambda}) = \boldsymbol{b} \Rightarrow (\boldsymbol{A}\boldsymbol{A}^\top)\boldsymbol{\lambda} = \boldsymbol{A}\boldsymbol{z} - \boldsymbol{b}.$$

$$\boldsymbol{\lambda} = (\boldsymbol{A}\boldsymbol{A}^\top)^{-1}(\boldsymbol{A}\boldsymbol{z} - \boldsymbol{b}).$$

Normal Cone to a Halfspace

Consider the halfspace

$$\Omega := \{y \in \mathbb{R}^n : \langle a, y \rangle \leq b\}, \qquad a \neq 0,$$

and a point $x \in \Omega$.

Interior point: If $\langle a, x \rangle < b$, then

$$\mathcal{N}_\Omega(x) = \{0\}.$$

Boundary point: If $\langle a, x \rangle = b$, then

$$\mathcal{N}_\Omega(x) = \{\lambda a : \lambda \geq 0\}.$$

Normal Cone: Two Halfspaces

Let

$$\Omega = \{y \in \mathbb{R}^n : \langle a_1, y\rangle \leq b_1,\ \langle a_2, y\rangle \leq b_2\}, \quad a_1, a_2 \neq 0.$$

If x lies at the intersection of the two boundaries,

$$\langle a_1, x\rangle = b_1, \qquad \langle a_2, x\rangle = b_2,$$

then

$$\mathcal{N}_\Omega(x) = \{\lambda_1 a_1 + \lambda_2 a_2 : \lambda_1, \lambda_2 \geq 0\}.$$

Interpretation: the normal cone is the conic hull of a_1 and a_2.

Normal Cone for Polyhedral Constraints

Theorem 7

Let

$$\Omega = \{x \in \mathbb{R}^n : \langle a_j, x\rangle \leq b_j,\ j = 1, \ldots, m\}.$$

For any $x \in \Omega$, the normal cone is

$$\mathcal{N}_\Omega(x) = \left\{ \sum_{j \in I(x)} \lambda_j a_j : \lambda_j \geq 0 \right\},$$

where

$$I(x) = \{j : \langle a_j, x\rangle = b_j\}.$$

Active constraints: indices in $I(x)$.

Equality Constraints

The constraint

$$\langle a, x \rangle = 0$$

is equivalent to the intersection of two halfspaces

$$\langle a, x \rangle \leq 0 \quad \text{and} \quad \langle -a, x \rangle \leq 0.$$

Both constraints are active at any feasible point x.

Normal Cone for Equality Constraints

From the polyhedral normal cone formula,

$$\mathcal{N}_\Omega(x) = \{\lambda_1 a + \lambda_2(-a) : \lambda_1, \lambda_2 \geq 0\}.$$

Therefore,

$$\mathcal{N}_\Omega(x) = \{(\lambda_1 - \lambda_2)a\} = \{\lambda a : \lambda \in \mathbb{R}\} = \mathrm{span}(a).$$

Complementary Slackness

For

$$\Omega = \{x : \langle a_j, x\rangle \leq b_j,\ j = 1, \ldots, m\},$$

the normal cone can be written as

$$\mathcal{N}_\Omega(x) = \left\{ \sum_{j=1}^{m} \lambda_j a_j : \lambda_j \geq 0,\ \lambda_j = 0 \text{ if } \langle a_j, x\rangle < b_j \right\}.$$

Since $\lambda_j \geq 0$, this condition is equivalent to

$$\sum_{j=1}^{m} \lambda_j \big(b_j - \langle a_j, x\rangle\big) = 0.$$

Complementary slackness

Matrix Form

Let

$$\Omega = \{x : Ax \leq b\}.$$

Then the normal cone can be written as

$$\mathcal{N}_\Omega(x) = \{A^\top \lambda : \lambda \geq 0,\ \lambda^\top (b - Ax) = 0\}.$$

Only active constraints contribute to the normal cone.

Linear Program

Consider

$$\max_x \; c^\top x \quad \text{s.t.} \; Ax \leq b.$$

Let

$$\Omega = \{x \in \mathbb{R}^n : Ax \leq b\}.$$

At optimality,

$$\nabla f(x^\star) \in \mathcal{N}_\Omega(x^\star).$$

Since $f(x) = c^\top x$,

$$c \in \mathcal{N}_\Omega(x^\star).$$

Normal Cone Characterization

For $\Omega = \{x : Ax \leq b\}$,

$$\mathcal{N}_\Omega(x) = \{A^\top \lambda : \lambda \geq 0,\ \lambda^\top (b - Ax) = 0\}.$$

Thus there exists $\lambda^\star$ such that

$$c = A^\top \lambda^\star, \qquad \lambda^\star \geq 0, \qquad (\lambda^\star)^\top (b - Ax^\star) = 0.$$

Dual Problem

Define the dual problem

$$\min_{\lambda} \; b^\top \lambda \quad \text{s.t.} \; A^\top \lambda = c, \; \lambda \geq 0.$$

Using complementary slackness,

$$(\lambda^\star)^\top b = (\lambda^\star)^\top A x^\star = (x^\star)^\top A^\top \lambda^\star = c^\top x^\star.$$

Strong duality

$$c^\top x^\star = b^\top \lambda^\star.$$